1 The Group 17 elements are oxidising agents.

(a) **(i)** Explain how the Group 17 elements act as oxidising agents.

.....

..... [1]

(iii) A student heats equal amounts of $I_2(g)$ and $H_2(g)$ in a sealed flask. The student leaves the contents to cool.

State what you would observe during the reaction.

..... [1]

(b) Cl_2 and Br_2 can each react with NH_3 to give N_2 and a hydrogen halide, HX.

$$3X_2 + 2NH_3 \rightarrow N_2 + 6HX \qquad X = Cl \text{ or } Br$$

The relative bond strengths of X–X and H–X determine the difference in enthalpy change of the two reactions.

(i) Describe and explain the difference in the X–X bond strengths of Cl_2 and Br_2.

.....

.....

.....

..... [2]

(ii) Describe the relative thermal stabilities of HCl and HBr.

..... [1]

(c) Table 1.2 shows some information about reactions of NaCl, NaBr and NaI.

Table 1.2

	NaCl	**NaBr**	**NaI**
observation with $Ag^+(aq)$	white precipitate		
type of reaction with concentrated H_2SO_4	acid–base	acid–base, then redox	acid–base, then redox
observations with concentrated H_2SO_4			• black solid • yellow solid • effervescence

(i) Complete Table 1.2. [4]

(ii) Suggest an identity for the species that produces each observation in the reaction of NaI with concentrated H_2SO_4.

black solid .....

yellow solid .....

effervescence .....

[2]

3 $H_2(g)$ and $I_2(g)$ react to form HI(g) in a reversible reaction, as shown in equation 1.

equation 1 $\quad H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$

In three separate experiments, a student combines different amounts of two or more of the gases from equation 1. In each experiment, the gases are left to reach equilibrium at a given temperature.

(a) In experiment 1, $H_2(g)$ and $I_2(g)$ are combined.

Water is then added to the equilibrium mixture to produce 1.00 dm^3 of solution **A**.

The amount of $I_2(aq)$ present in 1.00 dm^3 of solution **A** is found by titration with $Na_2S_2O_3(aq)$.

Exactly 32.90 cm^3 of 0.200 mol dm^{-3} $Na_2S_2O_3$ reacts with all of the I_2 in a 25.0 cm^3 sample of solution **A**.

$$I_2(aq) + 2S_2O_3^{2-}(aq) \rightarrow 2I^-(aq) + S_4O_6^{2-}(aq)$$

(i) Calculate the amount, in mol, of $S_2O_3^{2-}$ that reacts in the titration.

amount of $S_2O_3^{2-}$ = ... mol [1]

(c) In experiment 3, 0.0772 mol of H_2 and 0.0986 mol of I_2 are present in an equilibrium mixture at 298 K.

$$H_2(g) + I_2(g) \rightleftharpoons 2HI(g) \qquad K_c = 7.94 \times 10^2$$

Use the equilibrium constant, K_c, to calculate the amount, in mol, of HI present in the equilibrium mixture in experiment 3.

amount of HI in equilibrium mixture = ... mol [1]

(d) The value of K_c for the dissociation of HC*l* at 298 K is 5.50×10^{-34}.

$$2HCl(g) \rightleftharpoons H_2(g) + Cl_2(g) \qquad K_c = 5.50 \times 10^{-34}$$

(i) Define covalent bond.

.....

..... [1]

(ii) Describe and explain the relative thermal stabilities of the hydrogen halides HC*l* and HI.

.....

.....

..... [2]

(iii) Use the data given in **(c)** and **(d)** to suggest a value for the equilibrium constant for the **dissociation** of HBr(g) at 298 K.

..... [1]

6 **(a)** HOCl reacts with CH_2=CH_2 to produce $HOCH_2CH_2Cl$ in an electrophilic addition reaction.

(i) Define addition reaction.

.....

..... [1]

(ii) Describe how Cl_2 is used to produce HOCl.

.....

..... [1]

(iii) State **one** use for HOCl.

.....

..... [1]

3 The halogens chlorine, bromine and iodine show trends in chemical and physical properties down the group.

Table 3.1 shows some properties of chlorine, bromine and iodine.

Table 3.1

property	chlorine	bromine	iodine
colour and state at room temperature	green gas		
bond energy / kJ mol^{-1}	242	193	151
electronegativity	3.0	2.8	2.5
formula of sodium halide	NaCl	NaBr	NaI

(a) (i) Complete Table 3.1. [1]

(ii) The bond energy values in Table 3.1 refer to the X—X bond where X is the halogen.

Explain the trend in the bond strength of the X—X bond in the halogens.

.....

.....

..... [2]

(b) Explain, with the use of an equation, how chlorine, Cl_2, is used in water purification. State the role of the active species produced.

.....

.....

..... [2]

(c) The sodium halides in Table 3.1 also show trends in chemical properties.

(i) Identify the sodium halide that reacts with concentrated H_2SO_4 to form H_2S.

..... [1]

(ii) Identify the sodium halide that **only** undergoes a Brønsted–Lowry acid–base reaction with concentrated H_2SO_4.

..... [1]

(iii) A student adds a few drops of $AgNO_3$(aq) to a solution of NaBr(aq). State what is observed.

.....

..... [1]

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2 Chlorine is one of the elements in Group 17 of the Periodic Table.

(a) (i) Describe the colours of the Group 17 elements, chlorine to iodine, at room temperature.

.....

..... [1]

(ii) Describe the relative reactivity of the elements chlorine to iodine as oxidising agents.

..... [1]

(iii) State what is observed when chlorine reacts with hydrogen.

..... [1]

(ii) State the essential condition for chlorine to react with C_2H_6 at room temperature.

..... [1]

(iii) Write **two** equations to show the propagation steps in this reaction.

1 .....

2 .....

[2]

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.....

..... [1]

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..... [1]

(iii) State what is observed when chlorine reacts with hydrogen.

..... [1]

(iv) Explain why the thermal stability of the hydrogen halides decreases down the group.

.....

..... [1]

(ii) State the essential condition for chlorine to react with C_2H_6 at room temperature.

..... [1]

(c) $CHCl_3$ is another halogenoalkane. $CHCl_3$ forms when propanone reacts with NaClO.

NaClO is made from chlorine in a disproportionation reaction.

(i) Identify a reagent and conditions that can be used to convert chlorine to NaClO.

..... [1]

1 **(a)** The elements of Group 17 are called halogens.

Complete Table 1.1.

Table 1.1

halogen	colour at 293 K
chlorine	
bromine	
iodine	

[1]

(b) State the trend in volatility of the halogens chlorine, bromine and iodine. Explain your answer.

.....

.....

.....

..... [3]

(c) Iodine is made by reacting bromine with sodium iodide.

(i) Construct an ionic equation for the reaction of bromine with sodium iodide.

..... [1]

(ii) State the role of bromine in the reaction. Explain your answer.

.....

..... [1]

(d) Concentrated sulfuric acid is added to separate samples containing equal amounts of NaCl, NaBr and NaI. All three samples initially react to produce the hydrogen halide.

(i) Write an equation to describe the acid–base reaction that occurs when concentrated sulfuric acid reacts with NaBr.

..... [1]

(ii) Deduce which sodium halide, NaC*l*, NaBr or NaI, produces the largest percentage yield of hydrogen halide when concentrated sulfuric acid is added. Explain your answer by considering the relative reactivity of the halide ions as reducing agents.

identity of sodium halide .....

explanation .....

.....

.....

[3]

2 Chlorine, Cl_2, reacts with many elements and compounds to form chlorides.

Table 2.1 shows information about some chlorides of Period 3 elements.

Table 2.1

	Na	Mg	Si
formula of chloride			
structure of chloride	giant		
bonding of chloride			covalent
pH of solution formed on addition of chloride to water		6.2	

(a) Complete Table 2.1. [3]

(b) When Cl_2 reacts with **cold** NaOH(aq), Cl_2 is both oxidised and reduced. The products are NaCl, water and **G**.

(i) State the type of redox reaction in which the same species is both oxidised and reduced.

..... [1]

(ii) Identify **G**.

..... [1]

(iii) Write an equation for the reaction between Cl_2 and **hot** NaOH(aq).

..... [1]

(iv) Describe fully what is observed when $AgNO_3$(aq) is added to the aqueous solution of the chloride of sodium, followed by dilute NH_3(aq).

.....

..... [2]

1 The elements phosphorus, sulfur and chlorine are in Period 3 of the Periodic Table.

Table 1.1 shows some properties of the elements P to C*l*.

The first ionisation energy of S is **not** shown.

Table 1.1

property	P	S	C*l*
number of electrons in 3p subshell			
total number of unpaired electrons			
first ionisation energy /kJ mol^{-1}	1060		1260
formula of most common anion	P^{3-}	S^{2-}	Cl^{-}

(a) (i) Complete Table 1.1 to show the number of electrons in the 3p subshell and the total number of unpaired electrons in an atom of P, S and C*l*. [2]

(c) A student does three tests on separate samples of NaC*l*(aq).

Complete Table 1.2 with the observations the student makes in each test.

Table 1.2

test	test	observations
1	addition of a few drops of $Br_2(aq)$	
2	addition of a few drops of concentrated H_2SO_4	
3	addition of a few drops of dilute $AgNO_3(aq)$	

[3]

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3	addition of a few drops of dilute $AgNO_3(aq)$	

[3]

4 **(a)** Hydrogen chloride gas is made in the laboratory by adding concentrated sulfuric acid to potassium chloride.

(i) Construct an equation for this reaction.

..... [1]

(ii) Explain why hydrogen iodide is **not** prepared by adding concentrated sulfuric acid to sodium iodide.

.....

..... [2]

(b) A sample of HI(g) is added to a 2.00 dm^3 sealed vessel at 764 K and allowed to reach equilibrium.

reaction 1 $2HI(g) \rightleftharpoons H_2(g) + I_2(g)$ $K_c = 0.0217$ at 764 K

At equilibrium the mixture contains 1.70 mol of HI(g).

(i) State **one** difference in the appearance of the initial reaction mixture compared to the mixture at equilibrium.

.....

..... [1]

2 Chlorine is a reactive element. It forms many compounds.

(a) **(i)** Complete Table 2.1 to show the maximum oxidation number of the elements Na to P in their chlorides.

Table 2.1

element	Na	Mg	A*l*	Si	P
maximum oxidation number					

[1]

(d) **(i)** Write an equation for the reaction of chlorine with water.

..... [1]

(ii) Write an equation for the reaction of chlorine with hot NaOH(aq).

..... [1]

(e) Bleach is used as a cleaning product to kill bacteria. It is made by adding compounds like sodium chlorate(I), NaC*l*O, to water.

(i) Identify the formula of the ion present in bleach that kills bacteria.

..... [1]

3 Chlorine is a very reactive element.

(a) Chlorine reacts with silicon to form silicon(IV) chloride.
Describe the appearance of silicon(IV) chloride at room temperature and pressure. State its structure and bonding.

appearance .....

structure and bonding .....
[2]

(c) **(i)** State the reagent and conditions required for the formation of sodium chlorate(V) from Cl_2(g).

..... [1]

(ii) Explain why the reaction in **(c)(i)** is described as a disproportionation reaction.
Your answer should refer to relevant species and their oxidation numbers.

.....

..... [1]

2 The Group 2 elements Mg to Ba are all silvery-white reactive metals.

(a) (i) Draw a labelled diagram to show the bonding and structure of the Group 2 metals at room temperature.

[2]

(d) (i) Describe what is observed when $SrI_2(aq)$ reacts with concentrated sulfuric acid.

.....

..... [2]

(ii) Compound **X**, an anhydrous Group 2 bromide, is dissolved in water and titrated against aqueous silver nitrate.

A solution containing 0.250 g of **X** requires 33.65 cm^3 of 0.0500 mol dm^{-3} $AgNO_3(aq)$ for complete reaction.

Identify **X**.

Show your working.

X = ... [3]